Ganesh Gupta

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Experience

Oracle | *Software Engineer, AI Platform* Jun 2025 – Present

- Built **multi-agent system** via **OpenAI Agents SDK + MCP** tool-use handling **millions of daily transactions** at **99%+ uptime**.
- Designed and optimized **agentic AI** workflows across **GPT**, **Claude**, **Llama**, and **Gemini**, selecting and tuning models per use case for production **quality and cost**.
- Engineered pluggable **distributed executor** routing pipeline nodes to **Docker** or **AWS Lambda**, backed by **S3** artifact store eliminating redundant recomputation.
- Designed stateful recovery via per-node **Redis checkpointing**; interrupted workflows **resume from the last committed step** across **distributed execution**.

QureAI | *ML Infrastructure Engineer* Mar 2026 – Present

- Cut **p99 inference latency 420ms to 140ms** across **50K+ daily requests** migrating synchronous serving to async **vLLM** on **GCP Cloud Run** with autoscaling.
- Improved **RAG retrieval recall 61% to 89%** on a 2K-query eval set by rebuilding chunking with **hybrid dense-sparse retrieval** and **re-ranking**.
- Built **model observability** framework tracking **drift**, latency, and **eval regressions** in production, enabling governance-aligned **rollback** workflows.

Nomura Research Institute Financial Technologies | *Software Engineer* Sep 2019 – Aug 2023

- Built and operated **25+ production microservices** in **C++** and SQL-backed **distributed systems** for **1M+ users**, sustaining **99.9% uptime** over 4 years on **AWS** (EC2, RDS, SQS).
- Standardized API contracts and failure handling across **25+ services**, reducing escalations **60%**; drove **root cause analysis** and **post-incident reviews** to improve long-term reliability.

Projects

RootCause AI `github.com/ganeshhgupta/rootcause-ai`

- **LangGraph** multi-agent system diagnosing **production incidents** via tool-use chains; routes alerts to specialized sub-agents and synthesizes **root cause reports**, reducing **MTTR**.

Watch-your-LLM `github.com/ganeshhgupta/watch-your-llm`

- Production **LLM eval platform** with rubric-based scoring, **drift detection**, and latency tracking across model versions; **FastAPI**, **React/TypeScript**, **PostgreSQL**.

Distributed KV Store / VPC Control Plane Simulator `github.com/ganeshhgupta/distributed-kv-store`

- **Raft-consensus** KV store with **linearizable reads** and **log compaction**; companion **VPC control plane** simulator modeling subnet routing and security groups.

Education

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science* Jul 2016 – Jul 2020

Skills

Languages: Python, TypeScript, Java, Go, SQL

Agent Orchestration: OpenAI Agents SDK, LangGraph, MCP, Multi-Model Routing, Stateful Recovery

AI Infra: vLLM, RAG, Model Observability, AWS Bedrock

Infra: AWS, GCP, Docker, Kubernetes, Redis, PostgreSQL

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer